6 MAKING PROGRAMS EASY TO READ

Developers usually work in teams and it is important that they understand how each other's code and programs work, especially if there are errors that need correcting. Code should be written in standard ways and be explained using comments.

LEARNING OBJECTIVES

- Explain the benefit of producing programs that are easy to read
- Use techniques to improve the readability of code and describe how code works

CODE READABILITY

You should always try to ensure that any code you write is easy to read and understand. We refer to this as 'readability'. This benefits you and anyone else who needs to understand how your programs work.

It is surprising how quickly you forget. Try revisiting the programs you have already written in this unit and make sure it is still clear to you what they do and how they work. Imagine how much more difficult it would be to make sense of a complex program with lots of variables, subprograms, nested loops and multiple selection statements.

Programming is not a solo activity. Programmers usually work in teams, with each programmer developing a different part of the program. This only works if they all adopt a standard approach to writing readable code.

▶ **Figure 2.2** An example of Python code

```
Python 3.4.1: example2.py - C:/Python/example2.py
File  Edit  Format  Run  Options  Windows  Help
import random
game=True
while game==True:
    Num1=int(input("Choose one of the following: 1 Rock, 2 Paper, or 3 Scissors:"))
    Num2=random.randint(1,3)
    if Num2==1:
        print("Computer chooses Rock")
    elif Num2==2:
        print("Computer chooses Paper")
    else:
        print("Computer chooses Scissors")
    if Num1==Num2:
        print("It's a draw")
    elif Num1==1 and Num2==3:
        print("You win!")
    elif Num1==2 and Num2==1:
        print("You win!")
    elif Num1==3 and Num2==2:
        print("You win!")
    else:
        print("Computer wins")
    text=input("Play again? Y/N:")
    if text!="Y":
        game=False
```

GENERAL VOCABULARY

practice the way that you do something

SUBJECT VOCABULARY

block of code a grouping of two or more code statements

The programmer who produced the program in Figure 2.2 did not follow good **practice**. There are a number of ways that the readability of this code could be improved.

- Use descriptive names for variables (e.g. `userChoice` instead of `Num1`).
- Add blank lines between different **blocks of code** to make them stand out.
- Add comments that explain what each part of the code does.

HINT

Here's how to write comments in Python, Java and C#:

- **Python:** start each line of a comment with the # symbol
- **Java:** start single line comment with // or start a multi-line comment with /* and end it with */
- **C#:** Same as for Java.



▲ Programmers often work in teams; it is vital that any coding they share is clear and error free

Table 2.5 lists the techniques you should use to make your programs easy to read and understand.

DID YOU KNOW?

Microsoft Windows® uses roughly 50 million lines of code – code readability obviously becomes more and more important as programs get larger.

TECHNIQUE	DESCRIPTION
Comments	Comments should be used to explain what each part of the program does.
Descriptive names	Using descriptive identifiers for variables, constants and subprograms helps to make their purpose clear.
Indentation	Indentation makes it easier to see where each block of code starts and finishes. Getting the indentation wrong in Python will result in the program not running or not producing the expected outcomes.
White space	Adding blank lines between different blocks of code makes them stand out.

▲ **Table 2.5** Techniques for program clarity

SKILLS CRITICAL THINKING, PROBLEM SOLVING, CREATIVITY

ACTIVITY 11

REWRITING AND IMPLEMENTING ALGORITHMS

1 Program the following algorithm in a high-level language and make it readable by adding comments and indentation.

```
SET x TO 10
WHILE x >= 0 DO
 IF x > 0
  SEND x TO DISPLAY
 ELSE
  SEND 'Blast Off' TO DISPLAY
  END IF
 SET x TO x -1
END WHILE
```



▲ Calculators have a number of useful functions

2 Develop an algorithm for a simple calculator and code it in a high-level language that:
   - **a** allows the user to choose from these options: addition, subtraction, division and multiplication
   - **b** prompts the user to input two numbers
   - **c** performs the calculation and displays the result
   - **d** offers the user the option of performing another calculation.

SKILLS REASONING

SKILLS CRITICAL THINKING

SKILLS PROBLEM SOLVING

CHECKPOINT

Strengthen

S1 Why it is important to make your code easy to read?

S2 Outline the four techniques that a programmer should use to make code easy to read.

Challenge

C1 Revisit the programs you have already written. Do you still understand what they do and how they work? If not, try to improve their readability.

KEY POINT

Readability of code is important. You should always make your code easy to read.

How confident do you feel about your answers to these questions? If you're not sure you answered them well, look again at Table 2.5.

SUMMARY

- Using comments, descriptive names, indentation and white space makes code easier to read.
- Producing readable code makes it easier to understand what a program does and how it does it.